35 Kidney Lesions Associated With Dysproteinemias

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CHAPTER OUTLINE

KEY POINTS

- **Monoclonal gammopathy of renal significance is the clonal disorder that causes monoclonal immunoglobulin–induced kidney diseases that do not meet criteria for multiple myeloma or high-grade lymphoma.**
- **Monoclonal immunoglobulin causes kidney disease by deposition, precipitation, crystallization, and complement activation.**
- **Immunofluorescence demonstrating a monoclonal immunoglobulin is required to diagnose monoclonal gammopathy–associated kidney disease except in the cases of monoclonal gammopathy–related C3 glomerulopathy and monoclonal gammopathy associated thrombotic microangiopathy**
- **Treatment of monoclonal gammopathy of renal significance–related kidney diseases require clone-directed therapy.**
- **Only 20% to 30% of patients with proliferative glomerulonephritis with monoclonal immunoglobulin deposits have a detectable monoclonal immunoglobulin or clone.**

DYSPROTEINEMIA

Dysproteinemia indicates the presence of a monoclonal immunoglobulin. Monoclonal immunoglobulins are often produced by plasma cell clones and less commonly by B cell clones.[1] These patients are diagnosed with monoclonal gammopathy of undetermined significance (MGUS) if they have low tumor burden that does not meet criteria for multiple myeloma or high-grade lymphoma. The diagnosis is smoldering multiple myeloma, smoldering Waldenstrom macroglobulinemia, or low-grade lymphoma if tumor burden is sufficient but no end organ damage has occurred.[2] Patients with both the tumor burden and end organ damage are diagnosed with multiple myeloma, Waldenstrom macroglobulinemia, chronic lymphocytic leukemia, or high-grade lymphoma, and treatment should be promptly initiated. In the past, a problem would arise when patients with MGUS or smoldering diseases developed a kidney disease as a result of the monoclonal immunoglobulin. Due to the strict definition of kidney injury in multiple myeloma, which is limited to light chain cast nephropathy, and the lack of a renal definition in chronic lymphocytic leukemia or lymphoma, these patients were unable to receive appropriate treatment. Thus in 2012, the International Kidney and Monoclonal Gammopathy Research group introduced the term "monoclonal gammopathy of renal significance" (MGRS) as a novel hematologic etiology of kidney diseases and separated it from MGUS, other smoldering diseases, or low-grade lymphoma, which do not require treatment.[3]

MGRS is now recognized by the WHO as a pathologic clone that can come from the B cell or plasma cell line that produces a nephrotoxic monoclonal immunoglobulin but does not meet criteria for treatment based on tumor burden.[4] It is important to note that MGRS refers to the hematologic condition (the etiology) that caused the kidney disease but not the kidney disease itself. Treatment of MGRS requires clone-directed therapy determined by the nephropathic clone and not by the renal histology, as in traditional kidney diseases.[5] A number of kidney diseases have been commonly associated with MGRS and can be grouped by the presence or absence of monoclonal immunoglobulin deposits, whether the deposits are organized, and those with no deposits.[6] These lesions are reviewed in this chapter except for immunoglobulin-related amyloidosis, which is discussed in another chapter.

CLINICAL RELEVANCE

MGRS is a group of clonal hematologic disorders that result in kidney injury but do not satisfy the definition of multiple myeloma or malignant lymphoma. Many kidney diseases/lesions have been identified to be associated with MGRS. These kidney diseases are typically refractory to immunosuppressive therapies and have a high rate of relapse after kidney transplantation. Typically, clone-directed therapy is required to achieve a deep hematologic response, such as very good partial response or complete response in order to induce a renal response for preservation of kidney function. Successful treatment of the MGRS also significantly reduces the risk of recurrence after kidney transplantation.

MULTIPLE MYELOMA

Multiple myeloma is a malignant neoplasm of plasma cell origin. It is the second most common hematologic malignancy behind non-Hodgkin lymphoma. The diagnosis of multiple myeloma requires either >10% bone marrow plasma cells or >3 g/dL serum monoclonal protein and a myeloma-defining event (hypercalcemia, anemia, renal impairment, bone lesions) known as CRAB.[2] More recently, multiple myeloma can also be diagnosed with one of three biomarkers (>60% bone marrow plasma cell, involved to uninvolved free light chain ratio >100, >1 bone lesion on magnetic resonance imaging or positron emission tomography–computed tomography). Kidney disease is common in patients with multiple myeloma. Depending on the definition, 12% to 45% of the newly diagnosed patients have renal insufficiency with 6% to 8% requiring dialysis.

PATHOLOGY

The most common cause of acute kidney injury (AKI) in patients with multiple myeloma is cast nephropathy. Patients with light chain cast nephropathy typically present with AKI that is rapid in onset and can result in acute renal failure. This lesion is characterized by light chain–Tamm-Horsfall protein casts that obstruct post–loop of Henle tubules, resulting in acute tubular injury, inflammation, and subsequently fibrosis.[7] These casts appear fractured with sharp edges, stain PAS negative, and exhibit light chain restriction on immunofluorescence. The casts are often associated with mononuclear and giant cell reactions. Crystalline and amyloidogenic variants of light chain cast nephropathy exist. In addition, multiple myeloma patients can also have any of the kidney lesions that patients with MGRS may have but only light chain cast nephropathy is considered a myeloma-defining event. Patients with other kidney lesions cannot be diagnosed with multiple myeloma unless another myeloma-defining event (e.g., lytic bone lesions, hypercalcemia, and anemia) is present.[2] Finally, AKI can also result from plasma cell infiltration and extramedullary hematopoiesis as a result of overcrowding in the bone marrow.[7]

PATHOGENESIS

For light chain cast formation to occur, it requires a high serum-free light chain level (a minimum of 50 mg/dL of monoclonal free light chain, but the majority of patients have a serum concentration >150 mg/dL) and Bence-Jones proteinuria.[8] The Bence-Jones proteinuria is often > 200 mg/day.[9] The high concentration of monoclonal free light chain overwhelms the resorptive capability of the proximal tubules thus allowing large amounts of free light chain to enter the loop of Henle. There, the monoclonal free light chain binds and precipitate with Tamm Horsfall protein to form casts. It is important to note that light chain cast nephropathy can be precipitated by dehydration caused by hypercalcemia, vomiting and diarrhea, ACE inhibitors, and nonsteroidal antiinflammatory drugs, etc.[10]

TREATMENT AND OUTCOMES

Treatment required elimination of the potential causes and rapid reduction of the monoclonal free light chain concentration.[10] The monoclonal free light chain should be reduced by 50% to 60% rapidly and to <50 mg/dL by the end of cycle 1 of chemotherapy. Prompt initiation of chemotherapy is the key. Currently, a three-drug regimen containing bortezomib is the standard. The addition of daratumumab, an anti-CD38 antibody, has been shown to improve hematologic response and progression-free survival but has not been tested in patients with AKI. Extracorporeal methods of removing monoclonal light chains have been employed most notably with plasmapheresis and high cutoff dialyzers. Unfortunately, clinical trials have produced mixed results; thus neither modality has become standard therapy.[11,12] Patients who recover kidney function have a prognosis like patients who never developed AKI. Overall survival is inferior in patients with persistent renal dysfunction, especially those on dialysis.[13]

LESIONS WITH ORGANIZED IMMUNOGLOBULIN DEPOSITS

IMMUNOTACTOID GLOMERULOPATHY

Immunotactoid glomerulopathy (ITG) is a rare glomerular disease accounting for 0.06% of kidney biopsies.[14] Unlike other MGRS-associated kidney diseases in which plasma cell clones make up the majority of the pathologic clones, lymphocytic clones, particularly chronic lymphocytic leukemia clones, which include small lymphocytic lymphoma and some monoclonal B cell lymphocytosis, are the majority in ITG. One series from France found 60% of the clones were lymphocytic in origin with 48% belonging to the chronic lymphocytic leukemia/small lymphocytic lymphoma lineage.[15] Another series from the Mayo Clinic and Columbia University reported 76% of the clones were either lymphoma, chronic lymphocytic leukemia, or monoclonal B-cell lymphocytosis.[16] Multiple myeloma made up only 4% of the clones while 15% were MGRS. Not all cases of ITG are secondary to a monoclonal gammopathy. Polyclonal immunoglobulin deposits were found in 33% of cases in one series (polytypic variant of ITG).[16] Patients with polyclonal variants of ITG were more commonly found with an autoimmune disease (e.g., psoriasis, ulcerative colitis, systemic lupus erythematosus, rheumatoid arthritis, polymyositis, and multiple sclerosis) than in patients with the monoclonal variant (25% vs. 8%, $P = 0.07$). Positive antinuclear antibody was also more common in polyclonal than monoclonal ITG patients (35% vs. 17%, $P = 0.13$).[16] Less than half of the patients are hypocomplementemic (either low C3 or C4). Median age at presentation is 61 years with a slight male predominance (47%–67%). Patients with ITG present with heavy proteinuria typically >6 g/day, hypertension, and moderate renal impairment. Extrarenal manifestations have not been reported.

PATHOLOGY

ITG is defined by glomerular immunoglobulin deposits that organize into parallel microtubules with a distinct hollow core in the absence of a clinicopathologic diagnosis of cryoglobulinemic glomerulonephritis or lupus nephritis.[14,17] The glomerular pattern of injury on light microscopy (LM) is heterogenous and varies from predominantly membranous nephropathy (29%–59%) to membranoproliferative glomerulonephritis (MPGN) (29%–41%) to endocapillary proliferative GN (35%, Fig. 35.1A–35.1C).[14-16] It is not uncommon to see a mixture of these patterns in the same biopsy. Pure mesangial proliferative and diffuse sclerosing patterns are rare (<10%), whereas focal crescents are present in less than a quarter of cases.[16] Occasionally, large, glassy, PAS-positive, and silver-negative subendothelial immune deposits are seen. Cases associated with chronic lymphocytic leukemia/small lymphocytic lymphoma commonly exhibit direct interstitial infiltration by neoplastic B cells.[15,16]

On immunofluorescence (IF), most cases exhibit dominant staining for IgG with co-deposition of C3 (90%) and C1q (65%). However, exceptional cases of ITG of the IgA class or IgM class[18,19] and, more recently, a light chain–only variant of ITG have been reported.[20] Two-thirds of ITG cases exhibit immunoglobulin light chain isotype restriction (κ in two-thirds) with IgG subclass restriction (IgG1 in 67% and IgG2 in 27%) defining the monoclonal variant of ITG.[16] The remaining one-third of cases show polytypic deposits consistent with the polyclonal variant of ITG.[16]

On electron microscopy (EM), the deposits in ITG show microtubular substructure with a hollow center at magnification <50,000 and range in thickness from 10 to 60 nm. The deposits have well-defined borders and localize to the subepithelial zone (70%–100%), mesangium (84%), and/or subendothelial zone (75%). Some cases, especially those associated with chronic lymphocytic leukemia/small lymphocytic lymphoma, show smaller microtubules (<20 nm) comparable in size with fibrils of fibrillary glomerulonephritis (FGN). However, contrary to FGN, the microtubules do not permeate the mesangium or GBMs, and are DNAJB9 negative.[15,16] Intracytoplasmic microtubular inclusions can be seen in peripheral, bone marrow, and renal interstitial infiltrating chronic lymphocytic leukemia/small lymphocytic lymphoma cells.[15]

PATHOGENESIS

The pathogenesis of ITG is not fully understood. In monoclonal ITG, the deposits are composed of intact monoclonal immunoglobulin and complement components. The microtubular organization could be influenced by the molecular structure and physicochemical properties of the secreted monoclonal immunoglobulin, such as amino acid substitutions in the hypervariable region of the light chain or alterations in the charge and hydrophobicity of the variable regions of both heavy and light chains. While some investigators proposed that ITG could result from glomerular deposition of low-level circulating cryoglobulins, the vast majority of ITG patients do not have any serologic evidence or systemic manifestations of cryoglobulinemia. There are also notable pathologic differences between these diseases (e.g., contrary to cryoglobulinemic glomerulonephritis [CryoGN]), ITG cases do not exhibit abundant intracapillary infiltrating macrophages or intracapillary pseudothrombi), rendering this proposition unlikely.[15,16] The pathogenesis of the polyclonal variant could involve glomerular deposition of an unknown protein with intrinsic property of polymerization into microtubules, which then leads to autoimmune response. The IF findings of C3 and C1q glomerular deposition and glomerular proteomic findings of abundant peptide spectra for C3, C1q, C4, C5, C8, and C9[21] favor activation of the classical and terminal pathway of complement, which in turn triggers glomerular inflammation.

TREATMENT AND OUTCOME

In the largest series of ITG, rituximab-based therapy was used in 43% of monoclonal ITG versus 23% of polyclonal ITG patients. Another 32% of monoclonal ITG patients recieved chemotherapy versus 8% of polyclonal ITG patients. For the chronic lymphocytic leukemia and B cell clones, this was often accomplished with rituximab-based therapy. Approximately 25% of patients progressed to end-stage renal disease, which was more common in the polyclonal immunoglobulin deposits in patients (53% polyclonal vs. 11% monoclonal, $P < 0.01$).[16]

CRYOGLOBULINEMIC GLOMERULONEPHRITIS

CLINICAL CHARACTERISTICS

Of the three types of cryoglobulins, only two are involved with monoclonal gammopathy.[22] All type I cryoglobulins are composed of monoclonal immunoglobulin, which can be IgM, IgG, or IgA. Type II cryoglobulins are mixed, typically involving a monoclonal IgM κ (a rheumatoid factor) against polyclonal IgG, but not all type II cryoglobulins are the result of a lymphoproliferative or plasma cell proliferative disorder. The rest of the type II and type III cryoglobulins are

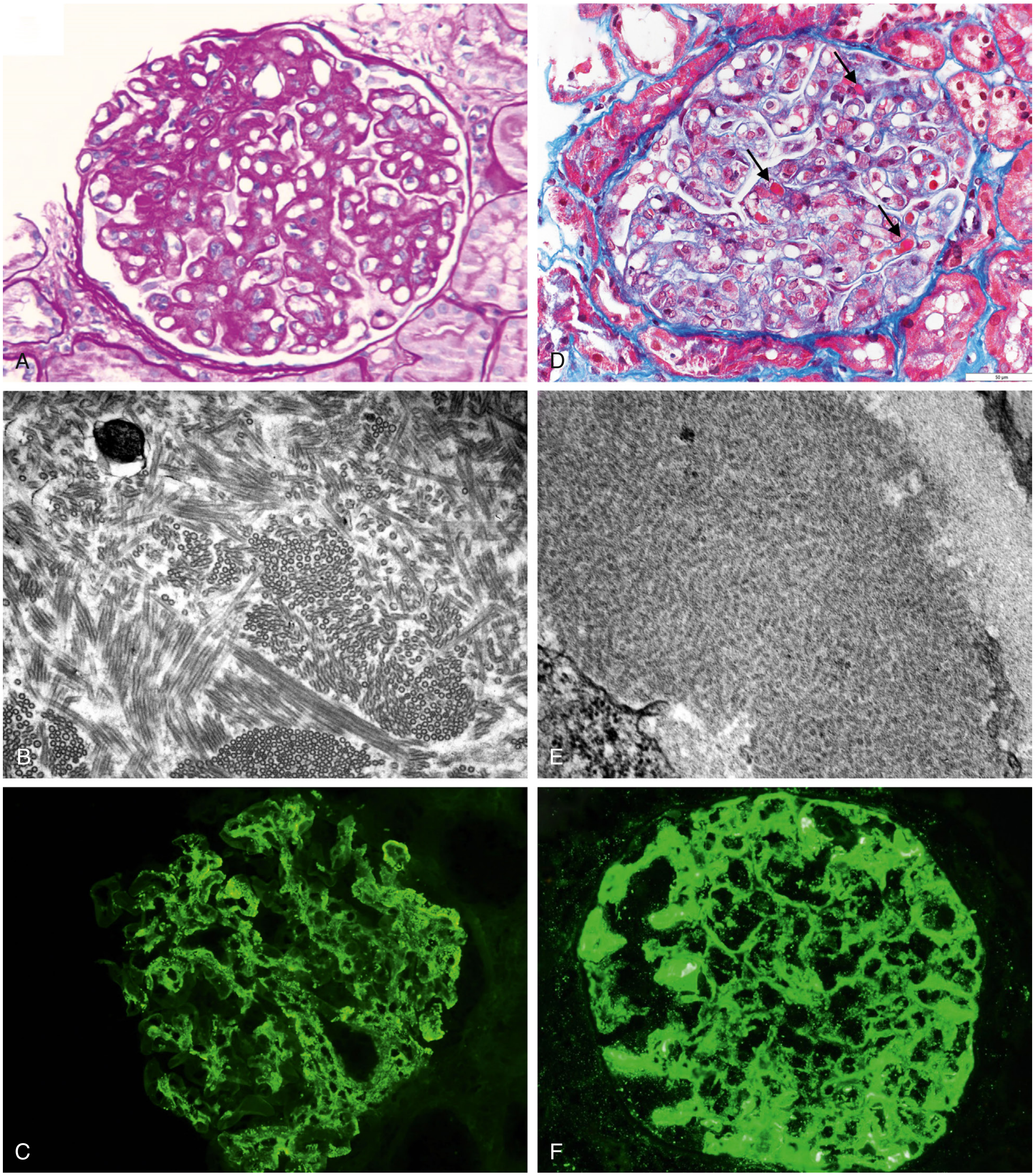

Fig. 35.1 Pathology of immunotactoid glomerulopathy and cryoglobulinemic glomerulonephritis type 2 associated with dysproteinemias. (A–C) Monotypic immunotactoid glomerulopathy. (A) A glomerulus exhibits a membranous nephropathy pattern of injury with diffuse thickening and vacuolization of the glomerular basement membrane. There is also mild mesangial hypercellularity (periodic acid–Schiff, ×400). (B) A high-power electron microscopy image shows the deposits to be composed of large microtubules with parallel alignment (×40,000). (C) On immunofluorescence, the glomerulus exhibits 3+ granular global mesangial and glomerular basement membrane staining for IgG (×400) and λ (not shown) with negative staining for κ (not shown). (D–F) Cryoglobulinemic glomerulonephritis type II. (D) Numerous macrophages infiltrating the glomerular capillaries. Trichrome-red intraluminal pseudothrombi *(arrows)* are present in some glomerular capillaries (×400). (E) On electron microscopy, some of the subendothelial deposits exhibit a substructure characterized by cylinders and short microtubules (×49,000). (F) On immunofluorescence, the glomerulus exhibits 3+ granular global mesangial and glomerular basement membrane staining for IgM (×400). The glomeruli also exhibited 2+ staining for IgG and κ with 1+ staining for λ (not shown).

usually secondary to infection, particularly from hepatitis C or autoimmune diseases. However, patients with hepatitis C who develop cryoglobulinemia do have a 35-fold increase in risk of developing non-Hodgkin lymphoma.[23] The median time for non-Hodgkin lymphoma development is 6.3 years. It is important to note that circulating cryoglobulins can be difficult to detect, so the absence of cryoglobulinemia does not rule out CryoGN.[24,25] Unlike ITG, extrarenal manifestations are common in CryoGN: hypertension; skin (rashes, purpura, digital necrosis); nerve (peripheral neuropathy); joint (arthralgia); vasomotor (Raynaud phenomenon); and others (abdominal pain, hemorrhage, and thrombosis).[22,26] Renal manifestations are variable with microscopic hematuria, and proteinuria (subnephrotic to nephrotic range) with or without renal impairment. Hypertension is usually severe. Renal failure is uncommon, occurring in 10% to 15% of patients.[27,28] Hypertension is not a risk factor for ESRD but is associated with cardiovascular death.

PATHOLOGY

LM in CryoGN exhibits macrophage-rich endocapillary proliferative GN or MPGN (see Fig. 35.1D–35.1F). Intracapillary pseudothrombi are seen in about 80% of cases. On IF in type I CryoGN, glomeruli stain for one immunoglobulin class (IgG or IgM) and one Ig light chain isotype (κ or λ). In type II CryoGN, there is usually glomerular staining for IgM, IgG, κ, and λ, with more intense staining for IgM than IgG and for κ than λ. On EM, the glomerular electron-dense deposits are subendothelial, mesangial, and intraluminal, whereas subepithelial deposits are uncommon. The deposits in three-quarters of cases exhibit organized substructure, most commonly short cylindrical/microtubular bodies. This substructure is more common in type II than type I CryoGN. CryoGN type I may show a variety of other substructures, such as lattice-like substructure or long straight microtubules with parallel alignment mimicking ITG.

PATHOGENESIS

The clones responsible for majority of the monoclonal cryoglobulinemia cases are of B cell origin, particularly the lymphoplasmacytic clone responsible for Waldenstrom macroglobulinemia, which accounts for 21% to 57% of cases in different series.[29-31] Plasma cell clones are most commonly responsible for type I cryoglobulinemia and are rare in type II cryoglobulinemia.[32]

TREATMENT AND OUTCOMES

Use of corticosteroids is an effective initial therapy, especially if life-threatening complications exist.[33] Plasmapheresis should also be considered in rapidly progressive cases. Steroid-sparing agents, such as alkylating agents like cyclophosphamide and rituximab, have a similar response rate of around 67% as frontline agents.[34] Clinicians should be cautious of cryoglobulin flares that occur if rituximab is used initially without an alkylating agent and which may also occur in patients with Waldenstrom macroglobulinemia. For plasma cell clones and lymphoplasmacytic lymphoma, bortezomib-based therapy can also be used. Relapse rates are high with 5- and 10-year event-free survival rates of 26.5% and 20.8%, respectively.[33] Patients with renal involvement and IgG type I cryoglobulinemia were at higher risk of relapse. Overall survival at 5 and 10 years were 77.0% and 52.5%, respectively. The most common causes of death were infection, hematologic progression, and heart failure.

CRYSTALGLOBULIN-INDUCED NEPHROPATHY

CLINICAL CHARACTERISTICS

One of the rarest and most severe complications of MGRS and MM is crystalglobulinemia. This entity is characterized by crystallization of monoclonal immunoglobulin intravascularly. These crystals can sometimes precipitate in colder temperatures, and the condition is referred to as cryocrystalglobulinemia.[35] Despite this property, cryoglobulin testing is often negative.[36,37] Kidney involvement in this syndrome is referred to as "crystalglobulin-induced nephropathy" (CIN).[38] Crystallization of monoclonal immunoglobulin in the renal microvasculature typically results in acute oliguric renal failure.[36] Most patients have mild proteinuria (range 0.6–1.1 g/day). Nephrotic syndrome is not a feature. Hematuria is present in just over half of patients. In the limbs, monoclonal immunoglobulin crystals can result in purpura, hemorrhagic bullae skin necrosis, and in severe cases digit infarct and gangrene.[37] Crystallization in large visceral vessels results in organ infarcts and often death.[35] Corneal and fundi deposits have also been reported.[39] One study found only 4 women out of 17 patients.[35] The median age was 55 years. Most of the patients had multiple myeloma, but more recent reports are in patients with MGRS.[36,37,39] In most patients, the diagnosis of CIN leads to discovery of an underlying plasma cell dyscrasia. Serum protein electrophoresis with immunofixation (SPEP/SIFE) is positive for M-spike in all patients, which is most commonly IgGκ and only rarely IgGλ or IgAk.[38,39] Urine protein electrophoresis with immunofixation (UPEP/UIFE) was positive for M-spike, and the serum free light chain (sFLC) ratio was abnormal in 50% and 56%, respectively, of reported patients with biopsy-confirmed CIN.[40,41] The nephropathic clone (typically plasma cell clone) is detected in the bone marrow in 70% of patients with CIN. While anemia, thrombocytopenia, elevated LDH, and low haptoglobin are frequent, peripheral smear schistocytes are not usually observed. Two-thirds of CIN patients have hypocomplementemia.

PATHOLOGY

Histologically, CIN is characterized by the presence of monoclonal immunoglobulin crystalline precipitates within the renal microvasculature (glomerular capillaries and vascular lumina), frequently causing occlusive thrombi, without associated immune-mediated glomerulonephritis.[38,41] The monoclonal immunoglobulin crystals appear hypereosinophilic on hematoxylin-eosin, trichrome-red, and silver-negative stains (Fig. 35.2E–35.2F). Occasionally, the crystals are accompanied by neutrophils, but glomeruli without crystals typically appear nonproliferative, without mesangial hypercellularity or leukocyte infiltration.[41] Secondary glomerular and/or

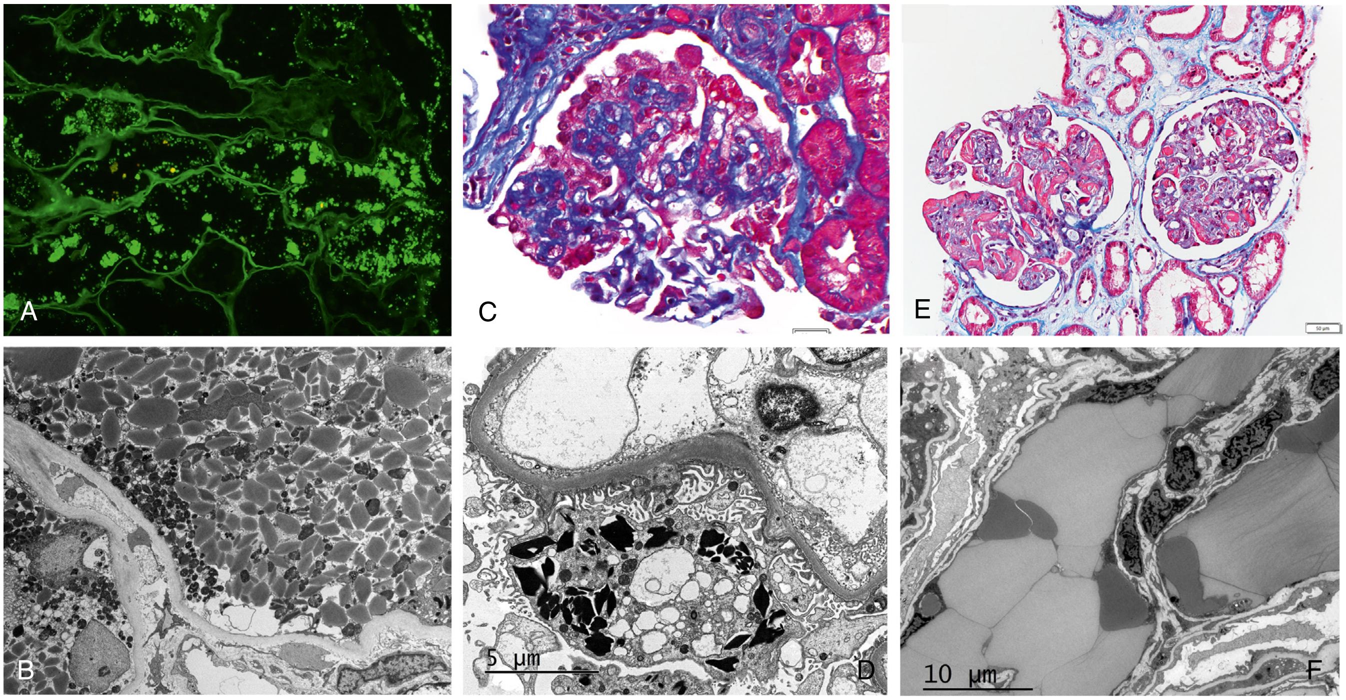

Fig. 35.2 Pathology of light chain proximal tubulopathy, light chain crystalline podocytopathy, and crystalglobulin-induced nephropathy. (A and B) Light chain proximal tubulopathy: (A) Immunofluorescence performed on paraffin tissue after pronase digestion reveals strong staining of proximal tubular cell crystals for κ (×400). The crystals were negative for λ (not shown). (B) Numerous rhomboidal and rod-shaped electron-dense crystals are present within proximal tubular cytoplasm (electron microscopy, ×2700). (C and D) Light chain crystalline podocytopathy. (C) The glomerulus exhibits collapsing features with retraction of the glomerular tuft and podocyte hypertrophy and hyperplasia (trichrome stain, ×400). (D) Highly electron-dense crystals with various geometric shapes are present in podocyte, within lysosomes, or free in the cytosol (×8000). (E and F) Crystalglobulin-induced nephropathy. (E) The glomerular capillaries are narrowed or occluded by numerous trichrome-red crystalline deposits (×200). (F) An electron microscopy image showing large mildly electron-dense crystalline deposits within glomerular capillaries (×5000).

vascular thrombosis as a reaction to the intraluminal crystal precipitation and endothelial cell injury is common, and thus CIN can be misdiagnosed as thrombotic microangiopathy (TMA). Similar crystals can be seen in distal tubule lumina, which are generally rare and not associated with cellular reactions. Tubules typically show acute injury.

On IF, the crystalline deposits most commonly stain with IgG and κ and only rarely with IgA and κ.[40] Analogous to light chain proximal tubulopathy (LCPT) and light chain crystalline podocytopathy, standard immunofluorescence on frozen tissue (IF-F) often fails to show the composition of crystals. In these cases, IF on paraffin tissue sections after antigen retrieval with a protease (IF-P), immunoperoxidase immunohistochemistry, or immunoEM can be revealing.

Ultrastructurally, the electron-dense crystals appear rectangular, rhomboidal, needle shaped, or rod shaped, and on high magnification they usually show parallel linear arrays with a reported thickness of 4 to 6 nm and periodicity of 9 to 20 nm.[38,40] Neither intracellular crystals nor mesangial, subendothelial, or subepithelial granular or organized electron-dense deposits are features of CIN, distinguishing it from crystal cryoglobulinemic glomerulonephritis type 1.

PATHOGENESIS

The pathomechanisms of crystalglobulinemia remain unknown. Intravascular crystallization may occur due to Fc–Fc interactions of the monoclonal immunoglobulin, possibly owing to abnormal glycosylation of the light chain portion of monoclonal immunoglobulin or the presence of unusual hydrophobic amino acids at the variable domain of immunoglobulin heavy or light chains, or through interactions with albumin.[42,43] The accumulation of crystalglobulins in the microvasculature potentially leads to endothelial cell injury and activation of coagulation cascade, resulting in thrombosis, occlusive changes, and subsequent ischemic injury to the kidney and other organs.[38]

TREATMENT AND PROGNOSIS

Prompt initiation of antiplasma cell therapy is crucial in the management of patients with crystalglobulinemia. Plasmapheresis can reduce the monoclonal immunoglobulin load and prevent further crystallization into microvasculature, and high-dose corticosteroids serve as bridging therapy until response to treatment of the underlying hematologic condition is achieved. Bortezomib-based regimens (e.g., CyBorD) appear to be effective in inducing complete or partial renal recovery and improvement of extrarenal manifestations in most patients when coupled with bridging plasmapheresis and high-dose corticosteroids.[39,44] While most patients treated with these regimens come off dialysis, relapses are common and could lead to irreversible kidney failure.[39,45]

LIGHT CHAIN PROXIMAL TUBULOPATHY

CLINICAL CHARACTERISTICS

LCPT is found in 1% of kidney biopsies.[46] It accounts for 0.5% of the kidney biopsies from multiple myeloma patients and 5% of patients with monoclonal gammopathies.[7,47] While LCPT is more common in patients with MGRS, the disease is more severe in patients with multiple myeloma.[48,49] Nearly two-thirds of the patients are male, and they typically present in their late 50s.[47,48] Clinically, LCPT presents with renal impairment usually indolent with low-grade proteinuria, although AKI can occur. Patients often have Fanconi syndrome characterized by potassium, phosphorus and uric acid wasting, normoglycemic glucosuria, and aminoaciduria.[49] Phosphorus wasting can result in osteomalacia and stress fractures. It is important to note that the electrolyte wasting often resolves as kidney function declines, but aminoaciduria will persist. The noncrystalline variant of LCPT is less likely to exhibit Fanconi syndrome.[50]

PATHOLOGY

Two morphologic variants of LCPT exist: a crystalline variant (86%) characterized by proximal tubular crystals and a noncrystalline variant (14%) characterized by light chain inclusions without crystal formation.[47,50] On LM, crystalline inclusions usually appear eosinophilic on hematoxylin-eosin and PAS-weak or negative, but they may appear optically clear. Acute tubular injury (ATI) is frequent, whereas interstitial inflammation is usually inconspicuous unless accompanied by light chain cast nephropathy.

With rare exceptions, crystalline LCPT is associated with κ light chain (Fig. 35.2 A–35.2B) while noncrystalline LCPT can be associated with either κ or λ light chains.[47,50] Importantly, due to the highly crystallized structure of monoclonal immunoglobulin light chain and intracellular localization, which could render the antigenic sites inaccessible to antibody binding, standard IF-F is insensitive (35%) for demonstrating light chain restriction of crystals. IF-P, which potentially denatures cell membranes and reveals the antigenic epitopes sequestered within the crystalline lattice, is much more sensitive (97%) and should be performed on cases with negative results on IF-F.[47,51] ImmunoEM is also sensitive, but this technique is only available in a few selected centers. In contrast, noncrystalline LCPT can readily be diagnosed on IF-F.[47]

On EM, proximal tubule crystals appear electron dense, needle shaped, rodlike, rhomboidal, or rectangular. They are present within phagolysosomes or lie free in the cytosol.[47] In noncrystalline LCPT, proximal tubular cells contain electron-dense droplets or vacuoles. In some cases, the cells appear markedly extended by intralysosomal indigested light chains (referred to as "lysosomal indigestion with constipation syndrome.")[46] A subset of cases of LCPT show concurrent monoclonal immunoglobulin lesions including light chain cast nephropathy, crystal-storing histiocytosis (CSH), light chain crystalline podocytopathy, AL amyloidosis, or light chain deposition disease.[47] CSH is somewhat similar to LCPT in that it is also characterized by light chain crystals, κ light chain predominance, and Fanconi syndrome. Crystals in LCPT are present predominantly in proximal tubular cells and limited to the kidneys, whereas crystals in CSH are predominantly present in interstitial histiocytes in the kidney and other tissues throughout the body including bone marrow and cornea leading to keratopathy.[52]

PATHOGENESIS

LCPT results from inability of the proximal tubule lysosomal system to degrade an overload of structurally abnormal filtered light chains. There is a notable homogeneity in the characteristics of light chains responsible for crystalline LCPT Fanconi syndrome. The majority belong to the Vκ1 variability subgroup and are derived from two germline gene segments, IGKV1-33 and IGKV1-39, and exhibit somatic hypermutations in complementarity-determining regions, which expose hydrophobic amino acid residues, making them resistant to degradation by lysosomal enzymes. This favors intralysosomal crystallization, lysosomal dysfunction, cell toxicity, and proximal tubule dysfunction.[53,54]

TREATMENT AND PROGNOSIS

LCPT has a relatively indolent course. ESRD occurs in approximately 20% of the patients and is more often in patients with AKI or multiple myeloma.[47] This, coupled with an observed low efficacy of conventional chemotherapy with alkylating agents and their frequent side effects (particularly myelotoxicity), led early reports to suggest deferring chemotherapy if there is no progression in the underlying hematologic condition or kidney dysfunction.[48] However, more recent studies using stem cell transplant and/or new generation of antimyeloma drugs (such as bortezomib-based therapy) demonstrated that achieving hematologic complete response (CR) or very good partial response (VGPR) is associated with stabilization or improvement of kidney function. In one study on LCPT Fanconi syndrome, ESRD developed in only 12% of treated patients versus 50% of untreated patients.[49] Thus effective chemotherapy could delay progression to ESRD in LCPT. Recurrence after kidney transplantation occurs rapidly within weeks to months.

LIGHT CHAIN CRYSTALLINE PODOCYTOPATHY

CLINICAL CHARACTERISTICS

Light chain crystalline podocytopathy is one of the newest kidney disorders recognized to be associated with MGRS. Most patients present with proteinuria (median 3.4 g/day) and CKD (median eGFR at presentation is 35 mL/min/1.73 m^2 but ranges from 4 to 113 mL/min/1.73 m^2, median serum creatinine at diagnosis is 1.9 mg/dL).[55] Full nephrotic syndrome is present in 28% to 35%.[55] Hematuria and HTN are present in 26% to 38% and 50% of patients. Overt Fanconi syndrome is uncommon (10%) and, when present, associated with concomitant LCPT. Extrarenal manifestation with crystalline keratopathy is reported in 22%.[55,56] The underlying hematologic condition in light chain crystalline podocytopathy is most commonly MGRS, less commonly symptomatic multiple myeloma, and only rarely marginal cell lymphoma, which is diagnosed concomitantly with light chain crystalline podocytopathy in most patients, whereas a minority of patients have a known history of monoclonal gammopathy before presentation with light chain crystalline podocytopathy. SPEP/SIFE and UPEP/UIFE are positive for M-spike in all patients, most commonly IgGκ and only rarely IgAk or IgGλ. sFLC is abnormal in 62% to 83% of patients (median dFLC 17 mg/dL).[55] The nephropathic plasma cell clone is detected in the bone marrow in 91% to 100% of patients (median % of monoclonal plasma cells 20%).

PATHOLOGY

On LM, two-thirds of cases exhibit focal segmental glomerulosclerosis (FSGS), which is of the collapsing variant in 67% (Fig. 35.2 C and 35.2D), non–otherwise specified (NOS) in 27% and tip in 7%.[55,57,58] The FSGS pattern is likely secondary to podocyte injury from the extensive intracytoplasmic crystalline inclusions. Misdiagnosis of light chain crystalline podocytopathy as primary or secondary FSGS is not infrequent if IF-P or EM is not performed.[58] In some cases glomeruli appear unremarkable on LM, whereas in others PAS-negative podocytes inclusions/crystals are visible and appear osmophilic on toluidine-blue stained semithin sections.

The defining feature of light chain crystalline podocytopathy is the presence of monoclonal immunoglobulin crystals within podocytes by EM, which are usually numerous and appear highly electron dense with various shapes (rod shaped, needle like, polygonal, rhomboidal, rectangular, or hexagonal) and membrane bound (intralysosomal) or free in the cytosol. On high magnification, they may exhibit a lamellated substructure characterized by parallel linear arrays with a reported periodicity of 4.4 nm.[55] In most light chain crystalline podocytopathy cases, similar crystals are seen in other renal cell types, most commonly within proximal tubule cells (80%, consistent with concomitant LCPT) and interstitial histiocytes (36%, consistent with concurrent CSH), and rarely in endothelial cells and mesangial cells.[55,59-61] Podocytes in most cases exhibit global foot process effacement. Podocyte crystals are of κ isotype in 92% and of λ in 8%. As in LCPT, IF-F is insensitive (12%) for demonstrating light chain restriction of podocyte crystals, whereas IF-P (77%), immunoperoxidase immunohistochemistry (83%), and immunoEM (100%) are more sensitive.

PATHOGENESIS

In light chain crystalline podocytopathy, the circulating free light chains potentially are endocytosed by podocytes and then crystallize inside lysosomes, leading to podocyte dysfunction, depletion, secondary FSGS, and albuminuria. Few studies have demonstrated that podocytes express megalin and CUBAM (cubilin-amnionless) complex (the receptors for albumin and light chain endocytosis by proximal tubule cells),[62] but this has not been confirmed by others. As in LCPT and CSH, light chains responsible for light chain crystalline podocytopathy are usually κ light chain, and the majority are derived from germline genes *IGKV1-33* and *IGKV1-39* and exhibit hydrophobic amino acid substitutions in positions 32 or 30, respectively,[55,63] which could render them resistant to degradation by lysosomal proteases.

TREATMENT AND PROGNOSIS

Mean ESRD-free survival time in patients with light chain crystalline podocytopathy is 58 months.[55] Kidney response to plasma cell–directed therapy (typically bortezomib-based therapy) depends on achieving hematologic response. In one study, kidney response occurred in 80% of patients who had hematologic CR or VGPR but in only 13% of those with hematologic partial response (PR) or no response (NR).[55] In this study, progression to ESRD was observed only in patients with FSGS (particularly the collapsing type). Thus early diagnosis and prompt initiation of plasma cell–targeted chemotherapy before development of FSGS is important.

FIBRILLARY GLOMERULONEPHRITIS

FGN, as the name suggests, is characterized by fibrillar deposits on EM. The fibrils are solid with a diameter (9–26 nm) that is larger than those of amyloidosis.[64] On LM, the most common pattern is mesangial proliferative glomerulonephritis. Historically, the deposits are defined as Congo-red negative; however, recent reports found a minority (4%) of FGN cases to be Congo-red positive.[65] Patients most often present with nephrotic range proteinuria and moderate renal impairment with a median creatinine of 2.1 mg/dL. It was initially included in the lesions associated with MGRS.[6] However, with the breakthrough finding of DnaJ homolog subfamily B member 9 (DNAJB9) that now defines fibrillary glomerulonephritis, it was discovered that the association DNAJB9+ cases with monoclonal gammopathy is exceeding low.[65,66] Thus with the exception of the exceedingly rare heavy chain variant,[67] FGN is no longer considered a kidney lesion commonly associated with MGRS.[68] No standard therapy currently exists for fibrillary glomerulonephritis, but the two most common agents used are cyclophosphamide and rituximab in non-MGRS-related cases.[64] Malignancy both solid and hematologic were found in 23% of patients while autoimmune diseases are present in 15%.

PATHOLOGY

On LM, the most common pathologic pattern is mesangial proliferative/sclerosing glomerulonephritis, followed by a membranoproliferative glomerulonephritis pattern.[64] Segmental membranous nephropathy features can be present in some cases. Endocapillary hypercellularity with leukocyte infiltration causing luminal occlusion has also been described, as well as crescents, which can occur in 17% of cases. On IF, most cases show strong smudgy staining for IgG while IgA and IgM can be seen in 28% and 47% of cases, respectively. C3 co-deposition is almost universally present, while C1q deposition is less common. Fibrils typically permeate the mesangial and lamina densa and subepithelial zone of the GBMs. These fibrils are solid and average around 18 nm in diameter. Fibrillary glomerulonephritis typically does not stain with Congo red, although a weakly Congo red–positive variant has been described.[65] Currently, the

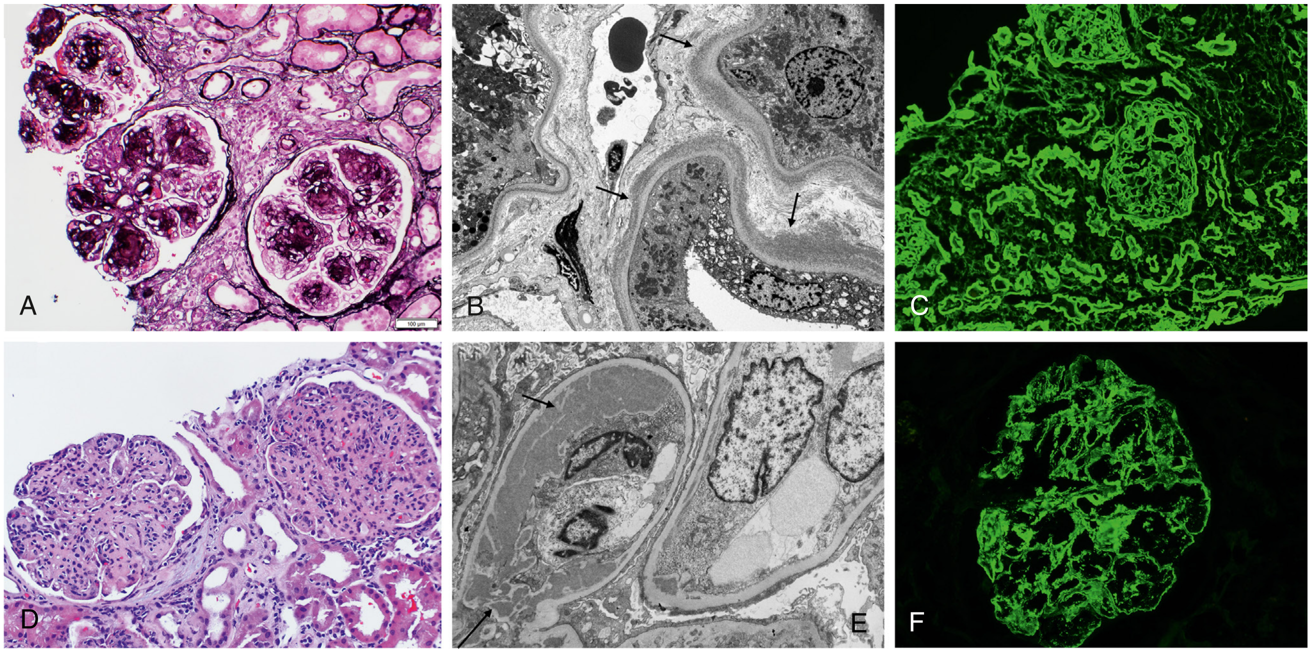

Fig. 35.3 Pathology of light chain deposition disease and proliferative glomerulonephritis with monoclonal immunoglobulin deposits. (A–C) Light chain deposition disease. (A) Glomeruli exhibit a nodular sclerosing pattern of injury reminiscent of nodular diabetic glomerulosclerosis (silver stain, ×200). (B) On electron microscopy, punctate-powdery, electron-dense deposits are present along the external aspect of the tubular basement membranes *(arrows)* (×4000). (C) There is diffuse linear staining of glomerular and tubular basement membranes for κ (×200) with negative staining for λ (not shown). (D–F) Proliferative glomerulonephritis with monoclonal immunoglobulin deposits. (D) The glomeruli exhibit a membranoproliferative pattern of injury with prominent mesangial hypercellularity and sclerosis and duplication of the glomerular basement membranes associated with cellular interposition (hematoxylin-eosin, ×200). (E) On electron microscopy, large granular electron-dense deposits (without substructure) are seen in the subendothelial zone *(short arrow)* and mesangium *(long arrow).* Podocytes exhibit diffuse foot process effacement (×4200). (F) There is bright granular mesangial and glomerular basement membrane staining for IgG3 (×400). Glomeruli were positive for IgG, κ, C3, and C1q and negative for IgA, IgM, λ, IgG1, IgG2, and IgG4 (not shown).

sine que non-characteristic of fibrillary glomerulonephritis characteristics is the positive staining for DNAJB9, which was first discovered at the Mayo Clinic.[69] When limiting fibrillary glomerulonephritis to positive staining with DNAJB9, the percentage of patients who has a MGRS dropped to <1% with the exception of heavy chain variant of fibrillary glomerulonephritis in which the IgG is monoclonal.[67,68]

PATHOGENESIS

The pathogenesis of fibrillary glomerulonephritis is not well understood. Whereas DNAJB9 has been proven to be an excellent tissue biomarker of the disease, a pathogenetic role in FGN and how it interacts with IgG remain unclear. In addition, how malignancy, autoimmune diseases, and hepatitis infection tie into DNAJB9 is also unanswered.[70]

LESIONS WITH NONORGANIZED IMMUNOGLOBULIN DEPOSITS

MONOCLONAL IMMUNOGLOBULIN DEPOSITION DISEASE

CLINICAL CHARACTERISTICS

Monoclonal immunoglobulin deposition disease (MIDD) is a group of diseases characterized by Randall-type deposits.[71] These deposits are noncongophilic and unorganized.[72] The deposits are composed of monoclonal light chain in light chain deposition disease (LCDD), monoclonal light and heavy chains in light heavy chain deposition disease (LHCDD), or truncated heavy chain in heavy chain deposition disease (HCDD).[73] The most common subtype is LCDD, which makes up >80% of cases.[73-75] Patients are mostly male (two-thirds) and most commonly present in their early 50s to mid-60s.[73-78] Proteinuria is usually in the nephrotic range with the heaviest proteinuria in patients with HCDD.[72,73,78] However, a nonproteinuric variant of MIDD has been described.[79] It is important to note that MIDD can exist with other lesions, most commonly with light chain cast nephropathy and immunoglobulin light chain (AL) amyloidosis.[76,80-82] Extrarenal manifestations can occur with MIDD but less common than in AL amyloidosis. The most common ones are cardiomyopathy and abnormal liver function tests, but peripheral nerves and gastrointestinal tract can also be involved.[72,83] Extrarenal manifestations appear to be more common in patients with symptomatic multiple myeloma than MGRS.[83]

PATHOLOGY

Nodular mesangial sclerosis is the most common feature on LM in MIDD, present in 50% to 60% of patients (Fig. 35.3 A–35.3C).[72,73,77,84] The nodules are reminiscent of Kimmelstiel-Wilson lesions and are most common in HCDD. The nodules and mesangial expansion are typically PAS positive, nonargyrophilic, and Congo-red negative. Mild mesangial hypercellularity can be present. Tubular basement membrane (TBM) thickening is often present and is much more common than GBM thickening. On IF, diffuse linear staining for the involved monoclonal immunoglobulin is always present along the TBM and frequently in the GBM and mesangium. Deposits are also commonly seen in arterial myocyte basement membranes. C3 and C1q deposits may be present mostly in cases of HCDD.

The deposits on EM are described as punctate, powdery, or finely granular and are deposited along the inner aspect of the GBM and outer aspect of the TBM. The deposits can be focal. Expanded mesangial areas by nonorganized deposits are more commonly seen in HCDD. MIDD is typically diagnosed by both IF and EM deposits. Cases of MIDD by IF only have been described, most commonly in cases with concurrent light chain cast nephropathy.[85] A small case series described the progression of IF-only MIDD to classic IF and EM MIDD in kidney allografts of patients with recurrent MIDD.[86] At the IF-only stage, these patients were asymptomatic, suggesting it is the early preclinical state of MIDD.

PATHOGENESIS

The pathologic monoclonal immunoglobulin is commonly produced by a plasma cell clone. The percentage of patients with multiple myeloma varies tremendously from 11% to 65%.[77,87] Using a more modern definition of multiple myeloma, the percentage of patients with multiple myeloma is typically between 20% and 34%.[72,88] Plasma cell MGRS clones make up 64% to 87% of the clones,[72,87] whereas lymphoplasmacytic clones make up 2% to 3% of the clones.[72,73,77] κ light chain restriction is more common, representing 81% of the cases in three series but was slightly less in two Italian series (57% and 68%).[72,73,76,77,87] κ light chain subtypes Vκ1 and Vκ4 are overrepresented in MIDD. Analysis of Vκ1 cases found mutations in the turn/loop region exposing hydrophobic amino acids, which may promote aggregation and precipitation.[89] For Vκ4, the mutations result in altered glycosylation, which again may promote aggregation and precipitation.[90] Furthermore, the variable domain of light chains responsible for LCDD is characterized by cationic complementarity-determining regions, which potentially promote their deposition on negatively charged GBMs and TBMs.[72] In HCDD, the immunoglobulin heavy chain is truncated at the first constant domain, resulting in the inability to bind immunoglobulin light chain. Additionally, free heavy chain variable domains in HCDD are cationic, favoring deposition along anionic sites along GBMs and TBMs.[91]

TREATMENT AND OUTCOMES

Treatment is dependent on whether the patient has symptomatic multiple myeloma or MGRS.[92] Patients with multiple myeloma should undergo standard treatment or be enrolled in a clinical trial. In patients with MIDD secondary to MGRS, limited clone-directed therapy may be effective.[5] For plasma cell clones, bortezomib and/or daratumumab, an anti-CD38 monoclonal antibody, have been the most effective.[87,93-96] Achieving a VGPR or better is associated with renal response and preservation of kidney function. For those who are unable to achieve a VGPR, an autologous stem cell transplantation is quite effective but is associated with higher morbidity.[87,96,97] Aside from achievement of VGPR, patients with eGFR >20 mL/min/1.73 m^2 at baseline are more likely to achieve a renal response.[88] Patients who develop ESRD are eligible for kidney transplant if they achieve a CR.[97,98] Otherwise, recurrent disease is frequent (up to 80%) and graft loss is common.

PROLIFERATIVE GLOMERULONEPHRITIS WITH MONOCLONAL IMMUNOGLOBULIN DEPOSITS

CLINICAL CHARACTERISTICS

Proliferative glomerulonephritis with monoclonal immunoglobulin deposits (PGNMID) is a recently described glomerular limited disease defined by glomerulonephritis and nonorganized deposits of monoisotypic immunoglobulin deposits.[99] The renal biopsy incidence is 0.21%,[99] but it is one of the most common MGRS lesions.[100] Patients typically present in their sixth to seventh decades of life with a median age at diagnosis of 56 years, but rarely adolescents and even children can be affected.[101,102] There is a slight female predominance (F:M ratio of 1.2:1). The majority of reported patients were Caucasian, but the disease occurs in Asians, African Americans, and Hispanics.[101] Most patients do not have clinical evidence of underlying infectious, autoimmune, or other systemic disease.[101] However, PGNMID has been rarely reported in patients with carcinoma; recent viral infection (hepatitis C, HIV, parvovirus B19); or autoimmune disease (e.g., autoimmune hemolytic anemia, rheumatoid arthritis, ankylosing spondylitis, and Sjögren syndrome).[102-104]

Patients usually present with proteinuria (median 3.8 g/day), hematuria (77%), and impaired kidney function.[101,103] About half of patients have full nephrotic syndrome. Two-thirds of patients have renal insufficiency at presentation (median eGFR 36 mL/min/1.73 m^2). Hypocomplementemia is present in about a quarter of patients (low C3 alone, low C4 alone, or low C3 and C4), with a higher frequency in the light chain–only variant (typically low C3 with normal C4).[101,103,105] Serum cryoglobulin and rheumatoid factor are typically negative.

The detection rate of the monoclonal immunoglobulin and bone marrow clone is lower (19%–37%) in the IgG, IgM, and IgA variants of PGNMID than other MGRS-related lesions.[101,103,106] Cases with IgG1 or IgG2 deposits are more likely to have a detectable monoclonal immunoglobulin than those with IgG3 deposits.[101] Using the standard sFLC ratio range, an abnormal sFLC ratio has a similar sensitivity to SIFE for the detection of paraprotein. UPEP/UIFE is less sensitive with a detection rate of 8% to 9%. Serum immunoblotting analysis and serum MALDI-TOF MS (matrix-assisted laser desorption/ionization-time-of-flight mass spectrometry) are more sensitive than SPEP/SIFE for detection of monoclonal immunoglobulin but are only available in few selected centers.[107] Hematologic malignancy is rare in PGNMID (3%–4% of patients) and includes chronic lymphocytic leukemia/small lymphocytic lymphoma and multiple myeloma.[101,103,106] The exception is the light chain–only variant of PGNMID, which is associated with a much higher detection rate of pathogenic plasma cell clone (73%) and serum monoclonal immunoglobulin (65%) with an abnormal sFLC ratio in 83% and association with symptomatic multiple myeloma in 29% of patients.[105]

PATHOLOGY

The glomerular alterations on LM in PGNMID are heterogeneous. The most common pattern, seen in up to two-thirds of cases, is MPGN (see Fig. 35.3 D–35.3F). The second most common pattern, seen in 20% to 35% of cases, is diffuse endocapillary proliferative glomerulonephritis.[101,103] A pure mesangioproliferative pattern is rare in the native kidney but is the most common pattern at initial diagnosis of recurrent PGNMID in the kidney allograft and is likely an early stage of disease.[108] A predominantly atypical membranous nephropathy pattern accounts for about 5% of cases.[109] Crescents are present in up to one-third of cases and tend to be focal.

By definition, IF reveals glomerular monoisotypic immunoglobulin deposits (positivity for a single immunoglobulin class, a single IgG subclass in the case of IgG class, and a single immunoglobulin light chain isotype). Contrary to MIDD, no TBM or vascular deposits are seen in PGNMID. The deposited immunoglobulin isotype in 90% of cases is IgG. In the remainder, it is IgM, IgA, or rarely (<1%) immunoglobulin light chain only.[101,103,105,110] Two-thirds of IgG isotype cases exhibit IgG3 subclass restriction, while IgG1 restriction is present in a quarter of cases.[101] There is C3 co-deposition in nearly all cases. C1q co-deposition is present in 55% to 64% of IgG cases but only 12% of immunoglobulin light chain–only cases.[101,103,105]

On EM, the deposits are present mainly in the mesangium and subendothelial zone, whereas subepithelial deposits are seen in 17% to 57% of patients. The electron-dense deposits exhibit a finely granular texture without substructure, resembling immune-complex type deposits.[101]

PATHOGENESIS

The pathogenesis of PGNMID is unknown but is likely mediated by the physicochemical and molecular characteristics of the monoisotypic immunoglobulin molecule. IgG3 is significantly overrepresented in PGNMID. This rare IgG subclass has several properties that allow it to be intrinsically "nephritogenic" including the highest molecular weight among the IgG subclasses (170,000 Dalton), making it more size restricted by the glomerular filtration barrier and the most positively charged subclass favoring affinity for intrinsic anionic sites in glomeruli. It has a strong tendency for self-aggregation promoting deposition in glomeruli and the greatest C1q complement fixing capacity triggering activation of downstream inflammatory mediators leading to glomerulonephritis.[101,111] IgG3 is unique in that it has a longer hinge region, the site of C1q activation and effector cell binding, than the other subclasses, accounting for its higher molecular weight and allowing for greater flexibility and increased ability of binding to complement and effector cells, rendering it a potent proinflammatory subclass. Importantly, since most patients with PGNMID of the intact immunoglobulin do not have a detectable circulating monoclonal immunoglobulin or a nephritogenic B cell or plasma cell clone and a small subset of cases occur following viral infections or in children, it has been hypothesized that at least some cases may not be monoclonal; rather, they are secondary to oligoclonal production of nephritogenic IgG3 due to a skewed B cell repertoire.[112] The pathogenesis of immunoglobulin light chain–only PGNMID is distinct from the other variants and is likely mediated by local activation of the alternative pathway of complement by the deposited monoclonal immunoglobulin.[105]

TREATMENT AND OUTCOME

Patients who present with subnephrotic proteinuria and normal kidney function may be treated initially with conservative therapy, and some of these patients remain stable for many years.[113] Patients who present with nephrotic proteinuria or impaired kidney function should be treated with clone direct therapy including rituximab-based therapy for cases driven by $CD20^+$ B cell clones and bortezomib-containing regiments for cases driven by plasma cell clones.[103,106] Management is much more challenging when no pathologic clone is identified, which is the majority of patients. Monotherapy with daratumumab has been recently found to decrease proteinuria and stabilize kidney function in most patients, but confirmatory studies are needed.[114]

The prognosis of PGNMID is variable. In one study in which follow-up (mean 30 months) was available on 32 patients, 38% had complete or partial kidney recovery, 38% had persistent renal dysfunction, and 22% progressed to ESRD.[101] Disease relapses are frequent in PGNMID, particularly in patients with detectable clones.[106] Recurrence of PGNMID after kidney transplant is common (87%) and occurs early post implantation (median time from implantation to diagnostic biopsy 5.5 months).[108] The prognosis of allograft PGNMID is guarded—44% of patients lose their allograft due to PGNMID within 3 years from diagnosis.[108]

LESIONS WITH NO IMMUNOGLOBULIN DEPOSITS

C3 GLOMERULOPATHY ASSOCIATED WITH MONOCLONAL GAMMOPATHY

CLINICAL CHARACTERISTICS

C3 glomerulopathy (C3G) is a recently described entity that is characterized by predominant C3 deposition without significant immunoglobulin deposits. C3G includes C3 glomerulonephritis (C3GN) and dense deposit disease (DDD), which can be differentiated by EM. Common causes include genetic or acquired complement alternative pathway defects, infection, and autoimmune disease, and an association with monoclonal gammopathy has been found, especially in patients older than the age of 50.[115-118] In patients with C3G older than the age of 50, 65% to 83% have a monoclonal gammopathy.[117,119] In fact, the reported median age at diagnosis for MGRS-associated C3G patients ranges from 54 to 65 years.[115-118] Patients often present with hypertension, renal impairment, nephrotic range proteinuria with or without nephrotic syndrome, and hematuria. Men make up 66% to 70% of the patients. Unlike younger patients with C3G, patients with MGRS-associated C3G are much less likely to have a complement gene mutation as compared with C3G not associated with monoclonal gammopathy (7% vs. 27%, respectively) in one study[120] and none in another.[121] Hypocomplementemia is seen in about 44% of patients similar to adult patients without monoclonal gammopathy.[120,121] In MGRS-associated C3G, roughly 20% of the cases are DDD.

PATHOLOGY

Two-thirds of the cases exhibit an MPGN pattern on LM while the mesangioproliferative pattern accounts for 21% to

22% of cases. [116,117,122] Crescentic glomerulonephritis and exudative glomerulonephritis patterns can also be present. On IF, bright C3 (2–3+) staining is seen in the mesangium and glomerular capillary walls while staining along the TBM is seen less frequently. It is important to point out that masked immunoglobulin deposits have been found in some cases of C3GN, which were revealed by paraffin IF after protease digestion.[123] These cases should be reclassified as MPGN with masked monoclonal immunoglobulin deposits. On EM, deposits in C3GN are seen in the mesangium, subendothelial, and/or subepithelial zone. DDD, on the other hand, is characterized by the presence of highly electron-dense, sausage-like, intramembranous deposits. Humplike subepithelial deposits, a classic finding in postinfectious glomerulonephritis, can be seen in both C3GN and DDD.

PATHOGENESIS

The monoclonal immunoglobulin in most patients with C3G with monoclonal gammopathy is a monoclonal IgG. IgA and IgM have been reported in about 20% of cases while monoclonal LC only is rare. The monoclonal immunoglobulin is produced by a plasma cell clone in 90% to 94% of cases, whereas B cell clones, particularly chronic lymphocytic leukemia clones, make up 6% to 10% of the clones. The hematologic condition is MGRS in 40% to 100% of cases.[115,117,118] While complement-activating monoclonal immunoglobulin has been found, interestingly, complement-activating polyclonal immunoglobulins have also been found in some patients.[120] Antifactor (F) H and anticomplement receptor 1 antibodies were the most common complement-activating antibodies (17% and 27%, respectively) in one study, while the C3 nephritic factor (nef) was most common in another study (45.8%).[120,122] Other antibodies include C4nef, anti-FI, and anti-FB.[120,122,124]

TREATMENT AND OUTCOMES

In patients with C3G without monoclonal gammopathy, treatment with mycophenolate mofetil and steroids results in a response rate between 67 – 79%.[125,126] The results are quite different in patients with MGRS. In a small study of 10 patients, 9 who were evaluable, 3 of 5 patients (2 of 3 treated conservatively and 1 of 2 treated with prednisone alone) treated with non-clone-directed therapy progressed.[115] Three of four (2 of 3 treated with cyclophosphamide-based therapy and 1 treated with bortezomib) patients treated with cytotoxic therapy responded. In another series of 23 patients, none of the 9 patients treated with immunosuppression (7 with mycophenolate mofetil and steroids, 1 with cyclophosphamide and 1 with eculizumab) had a hematologic response.[127] Kidney failure was observed in 33% of the patients regardless of therapy (conservatively, immunosuppression, or bortezomib-based clone-directed therapy) while preservation of kidney function was only observed in patients treated with immunosuppression (43%) and clone-directed therapy (57%). Therapy failure was more common in patients who were diagnosed with C3G in the kidney allograft, patients with a Columbia University C3G histologic index total chronicity score ≥ 4 ($p = 0.02$) and those without a hematologic response ($p = 0.03$). In the largest series to date of 50 patients with C3G and MGRS, patients treated with clone-directed therapy resulted in a renal response in 74% of patients vs 5% ($p = 0.001$) in those treated conservatively with ACE inhibition and immunosuppressive therapy consisting of steroids with mycophenolate mofetil, azathioprine, cyclophosphamide or rituximab.[118] Once again, hematologic response was associated with renal response but in this study, only patients who achieved a VGPR or better had a renal response ($p = 0.03$). More recently, daratumumab-based therapy has been reported to be beneficial in this disease although the number of patients studied is quite small.[93] For patients who progress to ESRD, kidney transplantation may be an option although recurrence is reported in 90% of patients and usually occurs within months if the monoclonal gammopathy is not eradicated.[128]

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH MONOCLONAL GAMMOPATHY

CLINICAL CHARACTERISTICS

TMA refers to two separate but related pathologic conditions.[129] The first is microvascular injury characterized by thrombosis in capillaries and/or arterioles with occlusive platelets, fragmented red blood cells, and fibrin thrombi accompanied by endothelial cell injury.[130] The second is microangiopathy hemolytic anemia (MAHA) characterized by hemolytic anemia (schistocytes, elevated lactate dehydrogenase, and low haptoglobin) and thrombocytopenia.[129,130] When the microvascular injury in the kidney occurs without evidence of MAHA, it is referred to as renal limited TMA. A recent study of biopsy-proven renal TMA revealed that 57% of the patients do not have signs of MAHA at the time of diagnosis.[129] The most common cause of renal limited TMA in that series was hematologic clonal disorders, of which only 18% of cases exhibited MAHA. The hematologic conditions associated with renal limited TMA include myeloproliferative neoplasms, plasma cell disorders, lymphoproliferative disorders, and lymphoplasmacytic disorders. Since myeloproliferative neoplasms are not associated with a monoclonal gammopathy, it is not discussed further in this chapter.[131] One multicenter study found that five of nine patients had multiple myeloma, one had Waldenstrom macroglobulinemia, and three had MGRS.[132] On the other hand, a single-center study of 20 patients found only 1 patient with multiple myeloma, 1 with Waldenstrom macroglobulinemia, and 1 with smoldering multiple myeloma, while 2 patients had POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy (nearly always λ), and skin changes).[133] The rest of the patients had MGUS. This is similar to a French series of 24 patients in which 2 had multiple myeloma, 2 had Waldenstrom macroglobulinemia, and 1 had chronic lymphocytic leukemia.[134] The median age of these patients is in the early to mid-60s.[132-134] AKI is the key feature with varying proteinuria from low grade (<1 g/day) to heavy (>10 g/day). In the French study, 71% of the patients required dialysis at the time of diagnosis but patients from this series were also more likely to have MAHA (91%) versus less than half of the patients from other series with less severe kidney injury.[132-134]

PATHOLOGY

TMA in the kidney can affect the glomerular compartment and/or the vascular compartment (arterioles and interlobular arteries).[129] On LM, glomerular involvement is characterized by the presence of intracapillary fibrin thrombi, fragmented erythrocytes, mesangiolysis, and/or duplication of the GBM, while vascular involvement is manifested as arterial or arteriolar thrombosis, mucoid intimal edema, and/or onion skinning.[129,132-134] Ischemic wrinkling of GBM is often present in cases with vascular involvement. On EM, widening of the subendothelial zone by electron lucent "fluffy" material is evident, with or without mesangiolysis and endothelial cell activation. Thrombi stain for fibrin by IF. Many cases of TMA associated with monoclonal gammopathy are subacute or chronic (i.e., without fibrin thrombi) and thus are best diagnosed by EM.

PATHOGENESIS

Although monoclonal immunoglobulin targeting the distal carboxyl terminus of ADAMTS13 (a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13) has been reported, thrombotic thrombocytopenic purpura as a cause of TMA with monoclonal gammopathy is uncommon.[135] In one study, only 20% of the patients tested had a low ADAMTS13.[133] Hypocomplementemia on the other is more common, seen in about 40% to 45% of patients.[132,133,136] A low C4 is more common than a low C3.[133] In a French series, 27% of patients had an anti-FH antibody but 20% of cases did not correlate with the monoclonal gammopathy and only one patient had CFHR1 deletion.[133] Pathologic complement variants were identified in 17% of patients including C3 and factor I variants. Further evidence of complement activation comes from a case where a patient had both DDD and TMA with MAHA.[137] In patients with POEMS syndrome, the endothelial injury is hypothesized to be secondary to cytokines (IL-6 and vascular endothelial growth factor) that are elevated in this condition.[138]

TREATMENT AND OUTCOMES

In patients who are not dialysis dependent, clone-directed therapy has been effective at resolving the TMA and preserving kidney function.[132] Plasma exchange should be started in conjunction with clone-directed therapy, especially in patients with severe AKI.[133] For patients who are on dialysis at presentation, only 16.6% of patients treated with clone-directed therapy with or without plasma exchange recovered kidney function compared with 57.1% of those who received eculizumab with or without plasma exchange.[134]

GLOMERULAR INVOLVEMENT BY WALDENSTROM MACROGLOBULINEMIA

Waldenstrom macroglobulinemia is defined by the presence of a circulating monoclonal IgM of any size and at least 10% lymphoplasmacytic lymphoma cells in the bone marrow with end organ damage most commonly from anemia, hyperviscosity, adenopathy or organomegaly, peripheral neuropathy, cryoglobulinemia, or cold agglutin disease.[139] The estimated cumulative incidence of Waldenstrom macroglobulinemia–related nephropathy is 5.1%, and kidney diseases related to Waldenstrom macroglobulinemia include immunoglobulin-related amyloidosis, CryoGN, LCDD, light chain cast nephropathy, thrombotic microangiopathy, and direct interstitial infiltration by lymphoplasmacytic lymphoma.[140]

A distinctive glomerular lesion in Waldenstrom macroglobulinemia is characterized by voluminous glomerular intracapillary monoclonal IgM thrombi. Macroglobulinemic nephropathy is the historical term used to describe this lesion.[141] However, since this lesion can be seen in other IgM-related disorders and B cell lymphomas, Audard and colleagues[142] proposed an alternative term, intracapillary monoclonal deposits disease (ICMDD), to describe this lesion. ICMDD is exceedingly rare today. In a recent study on kidney lesions associated with Waldenstrom macroglobulinemia and other IgM-producing lymphoproliferative disorders, immunoglobulin-related amyloidosis and CryoGN were the two most common lesions, 33% and 28%, respectively, whereas ICMDD accounted for only 4% of cases.[143] Patients with ICMDD typically present with AKI, nephrotic syndrome (60%, mean proteinuria of 6 g/day), and microhematuria (60%).[142] In patients with ICMDD, Waldenstrom macroglobulinemia–directed chemotherapy usually leads to recovery of kidney function and remission of nephrotic syndrome.[142]

ICMDD is characterized by massive, amorphous, PAS-positive, and silver-negative subendothelial and intracapillary deposits. Similar deposits can be seen in arterioles and interstitium.[142,143] Contrary to CryoGN, in ICMDD mesangial sclerosis and hypercellularity, endocapillary hypercellularity, intracapillary infiltrating macrophages, and full-blown MPGN pattern are lacking. On IF, the intraluminal immune deposits stain for IgM and either κ or λ light chains. On EM, they typically appear granular without substructure (in contrast to CryoGN).

It was initially thought that ICMDD results from precipitation of the circulating monoclonal IgM in glomerular capillary due to increased blood viscosity.[144] However, some patients with ICMDD do not have hyperviscosity syndrome and thus other contributing factors, such as the physicochemical properties of monoclonal IgM and activation of the classical pathway of complement, potentially play a role in pathogenesis.[142]

TREATMENT AND PROGNOSIS

Treatment depends on whether the patient has Waldenstrom macroglobulinemia or MGRS. In patients with ICMDD without Waldenstrom macroglobulinemia, rituximab ± steroids could be used. It is important to note that IgM/cryoglobulin flare could occur with rituximab monotherapy, and in severe cases, plasmapheresis and steroids are required before rituximab can be resumed.[145] In patients with Waldenstrom macroglobulinemia with hyperviscosity, plasmapheresis should be performed before initiating chemotherapy. Frontline therapy for Waldenstrom macroglobulinemia consists of either bendamustine and rituximab or a Bruton tyrosine kinase inhibitor such as ibrutinib or zanubrutinib.

CONCLUSION

Kidneys are among the most common targets of monoclonal immunoglobulins, which are produced by B cells and plasma cell clones. Although AKI is common in patients with multiple myeloma, these clones do not have to reach a malignant stage to cause kidney injury. Clones that cause kidney injury by their monoclonal immunoglobulin that do not meet criteria for the perspective cancer are now called MGRS. The diagnosis of MGRS-related kidney diseases requires a kidney biopsy, but the treatment depends on the nature of the clone that produced the monoclonal protein. This requires a full hematologic evaluation that may include bone marrow biopsy, peripheral blood flow cytometry, and positron emission tomography–based imaging studies. In addition, since immunosuppressive therapy has little success with B cell and plasma cell clones, cytotoxic therapies and immunotherapies are required. The management of these patients requires collaboration from the nephrologist, whose clinical suspicion leads to the biopsy; the pathologist, who makes the diagnosis by appropriate testing; and the hematologist, who treats the pathologic clone. Some patients in whom the disease is too advanced also require assistance from a nephrologist in terms of renal replacement therapy. This is truly a multidisciplinary disease that requires the collaboration and understanding of disease mechanisms from all parties to ensure a successful outcome.

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